# Ebrahim Alhaddad

**Contact:** efalhadd@gmail.com | **Cell:** (267) 261-2296 | **Social:** https://ebrahimalhaddad.github.io/quarto-website/

## PROFESSIONAL SUMMARY

Analytics-focused software engineer with nearly 4 years of experience across **Apple**, **Qualcomm,** and AI/backend environments, now completing an MSBA. Experienced in Python, model evaluation, search/retrieval workflows, and performance analysis, with a focus on turning messy data and system behavior into decision-ready insights

## SKILLS

**Analytics:** Python | Pandas | Polars | SQL | EDA | Feature Engineering | Metric Definition | A/B Testing | Model Evaluation | Data Visualization | Causal Inference | Segmentation | Thresholding & Targeting | Recommender Systems

**AI & Data:** LangChain | LangGraph | Vector Databases | RAG | LLM Evaluation | OpenSearch (Elasticsearch)

**Backend & Cloud:** FastAPI | REST APIs & SSE | Docker | CI/CD | AWS (CDK, ECS, ECR, Amplify, Bedrock)

**Diagnostics:** C/C++ Code Analysis | OS-level diagnostics | Performance Analysis | Log/Trace Analysis | Root-Cause Analysis | Lab Instrumentation

**Technical Delivery:** Prototyping | Cross-Functional Communication | Requirements Translation | Technical Documentation

**Certification:** IBM AI Engineering Professional Certificate **Languages (Spoken):** English, Arabic

## EDUCATION

***Master of Science in Business Analytics***, Rady School of Management — 06/2025 - 06/2026
University of California, San Diego — San Diego, CA
**Honors:** Crown Prince International Scholarship (2025 - 2026), Rady Fellowship (2025 - 2026)

***Bachelor of Science in Electrical Engineering*****,** Viterbi School of Engineering — 09/2015 - 12/2019
University of Southern California — Los Angeles, CA
**Honors:** Crown Prince International Scholarship (2013 - 2019), USC Dean's List (2015 - 2016)

## EXPERIENCE

### Rarible — 05/2025 - 07/2025

**Software Engineer — Contractor** — Remote

- Rebuilt a prototype into a deployed AI-enabled application using FastAPI, LangGraph, Redis, DynamoDB, and LLM APIs, adding workflow logging for performance and user-behavior analysis
- Designed backend workflows for chat, code execution, and asset generation, improving reliability and making application behavior easier to monitor
- Implemented data capture for demo campaigns, enabling analysis of customer behavior, system performance, and feature usage patterns
- Configured AWS deployment and CI/CD workflows using CDK, ECS/Amplify, and GitHub Actions, improving deployment repeatability

### Array Innovation (Fast-Growing AI Startup) — 05/2024 - 05/2025

**AI Engineer — Prototyping & Backend Systems** — Bahrain

- Built evaluation pipelines used by cross-functional teams to test AI/retrieval behavior across demos and project iterations, turning subjective output quality into repeatable metrics and driving a 20% retrieval-performance improvement
- Developed FastAPI microservices for retrieval, RAG, keyword extraction, fuzzy matching, and embedding-based workflows used across larger systems
- Analyzed retrieval failures and output-quality issues to identify data-quality gaps, guide improvements in data ingestion, and prototype features
- Drove technical delivery of prototypes, translating client requirements into API designs, workflows, and deployable proof-of-concept systems
- Partnered with business analysts, senior engineers, and client-facing teams to clarify vague requirements and shape feasible solution designs

### Apple — 08/2021 - 02/2023

**Software Engineer — Core Media IMG Team** — San Diego, CA

- Owned technical triage for 9,000+ issue reports across media components, detecting recurring failure patterns, duplicate clusters, and out-of-scope reports to improve defect prioritization and routing
- Synthesized crash reports, stackshots, performance traces, system logs, and reproduction data into technical findings that narrowed ambiguous failures into likely root causes
- Scripted Python automation for log parsing, trend extraction and tagging, reducing manual analysis and surfacing recurring failures
- Coordinated across media, performance and kernel teams to resolve ownership ambiguity, validate analysis and drive critical issues toward resolution
- Diagnosed memory, latency, power, and concurrency failures in a large C codebase, using trace/log evidence to support engineering decisions

### Qualcomm — 05/2020 - 08/2021

**Performance Engineer — (Machine Learning / Computer Vision) R&D Team** — San Diego, CA

- Automated lab instrumentation workflows in Python asyncio to capture synchronized power, bandwidth, and latency measurements across DSP, CPU, and cache resources, replacing manual timing and post-run cleanup
- Collected and analyzed performance data across workload and resource configurations to support SoC power modeling and KPI definition
- Analyzed raw traces and function-level measurements to identify compute-pipeline behavior, validate measurement quality, and ensure performance metrics were accurate and defensible
- Designed quantitative benchmarking protocols for feature performance across devices, translating ambiguous comparison goals into measurable tests
- Produced performance KPIs and presented findings to engineering leadership to support platform-design and roadmap decisions

## PROJECTS

### Analytics Consulting Capstone — Spring 2026

- Led technical discovery for a client-facing search relevance project, translating vague product issues into a roadmap for search-quality improvements
- Defined evaluation approach and success metrics for search quality, data usability, and client-facing recommendations
- Analyzed large-scale search, item, and vendor datasets to identify issues and limitations of search relevance and downstream analytics
- Prototyped search-improvement workflows using fuzzy matching, canonicalization, keyword analysis, and OpenSearch-based tuned retrieval